

Audio MCU Development Board Design Notes

Document status: v0.1 — initial repository design record

Project state: architecture selected; schematic and PCB are in development

Primary use: reusable real-time audio and guitar-effects development platform

1. Purpose and first-revision scope

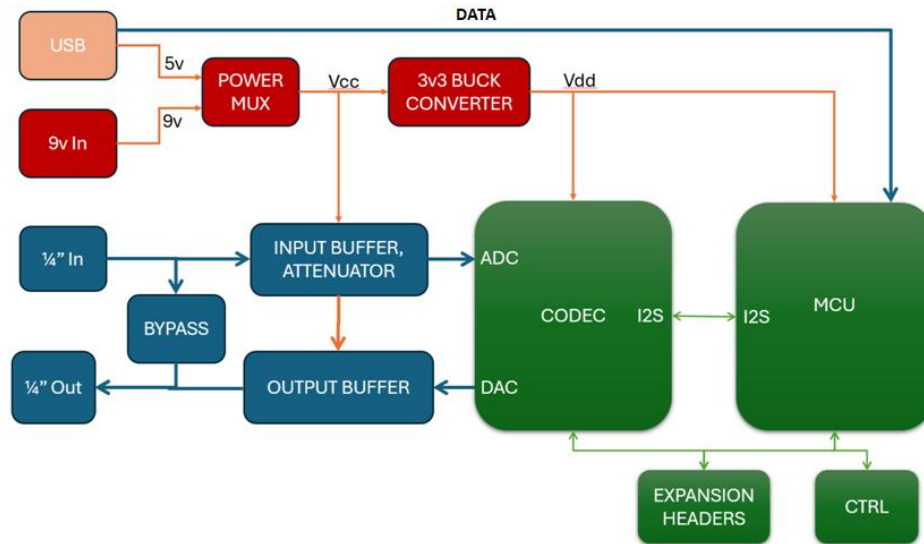
This board is a custom embedded platform for learning and experimenting with real-time digital audio processing. It accepts a guitar or other instrument-level signal, conditions and digitizes it with an external audio codec, processes the audio on an STM32 microcontroller, and reconstructs an analog output signal. The board is intended to make DSP firmware experiments repeatable without rebuilding the analog, power, and programming infrastructure each time.

Revision A is deliberately a development board, not a finished production guitar pedal. The initial hardware goal is reliable power, programming/debug, codec bring-up, analog I/O, and basic physical controls. Mechanical enclosure design, polished pedal UX, USB-audio support, and feature-complete effects firmware are outside the initial bring-up target.

Area	Initial target
Audio input	One 1/4 in TS instrument input with approximately 1 Mohm input impedance
Audio output	One 1/4 in TS low-impedance line/instrument-level output
Audio format	24-bit, 48 kHz baseline; stereo-capable hardware with mono operation first
Processing	Real-time STM32 DSP, initially a pass-through and simple effects chain
Codec link	SAI/I2S audio data plus I2C configuration/control
Power	9 V DC pedal supply and USB-C 5 V input, with controlled power-source selection
Development access	USB-C device connection and dedicated SWD programming/debug
Physical controls	Four potentiometer inputs, at least two digital inputs, and at least three LEDs
Expansion	Accessible GPIO and I2C, SPI, and UART expansion points
PCB	Four-layer board preferred

2. Selected architecture

The core architecture is an STM32H573 MCU with a TLV320AIC3204 external audio codec. This separates real-time DSP from analog conversion and is the right trade-off for a reusable audio platform: the codec supplies the audio-oriented ADC/DAC and analog routing, while the MCU is free to run the effect algorithm and user-interface logic.



The STM32H573 family supports two SAI peripherals, USB 2.0 full-speed host/device capability, and SWD/JTAG debug support. The TLV320AIC3204 provides the audio serial interface, control interface, ADC/DAC functions, and analog routing needed for a compact audio front end. The codec has MCLK, BCLK, WCLK, DIN, and DOUT connections for its primary serial audio interface.

2.1 Clocking and digital-audio ownership

The STM32 will initially be the audio-clock master and the codec will operate as the audio serial slave. This gives the firmware one clock tree to manage and makes the first pass-through implementation easier to reason about.

Baseline format:

- 48 kHz sample rate
- 24-bit samples, carried in a practical SAI/I2S frame size selected during firmware configuration
- Full-duplex receive/transmit using DMA ping-pong buffers
- Codec configured over I2C; its reset pin driven by the MCU

At 48 kHz, a conventional $256 \times f_s$ master clock is 12.288 MHz. The exact MCLK source, PLL settings, SAI frame format, and pin mapping must be verified against the selected STM32 package and the codec clocking tables before the schematic is frozen.

2.2 Firmware model

The intended firmware structure separates hard real-time audio work from slower control work:

1. DMA transfers a completed block of samples to an audio callback.
2. The callback processes the block through a modular DSP chain and returns the output block.
3. A slower foreground/control loop reads potentiometers and switches, updates parameters, manages LEDs, and handles USB or diagnostic messages.

The first firmware milestone is codec initialization followed by an audibly clean, unity-gain pass-through. Effects should be added only after measured latency, noise, level headroom, and buffer stability are understood.

3. Analog audio path

3.1 Instrument input

Passive guitar pickups expect a high input impedance. The input stage should therefore present about 1 Mohm to the jack, while protecting the rest of the system from connector ESD, cable transients, and accidental over-level input.

Important design decisions to document on the schematic:

- Clamp parts must have low enough capacitance that they do not noticeably load or dull a passive pickup.
- The bias point must be quiet and appropriate for the codec's allowed analog-input common-mode range.
- The buffer must operate cleanly from the chosen analog supply and have adequate input/output headroom for expected guitar and boosted-pedal levels.
- The ADC anti-alias network is a modest analog filter; it complements, rather than replaces, the codec's internal conversion filtering.
- Gain staging must leave practical headroom. "Unity" at the jack, analog front end, codec, and DSP full-scale are not automatically the same voltage.

The initial board will use only one input channel, but the codec connection should preserve a clean path to a second channel or alternate input-routing experiment later.

3.2 Output path

The codec DAC output is not assumed to be a finished pedal output by itself. The output stage should provide the required filtering, DC blocking or bias removal as appropriate, a robust low output impedance, and protection for a 1/4 in TS jack.

The first revision should make output level configurable or easy to alter with resistor options. That will let the board be evaluated as a line-level development output first and tuned later for the desired pedal/instrument interaction.

3.3 Grounding and layout intent

Audio quality will be determined as much by layout as by the block diagram. The board should keep the high-impedance input node short and away from USB, switching nodes, MCLK/BCLK traces, and LED currents. The codec analog supply filtering, local decoupling, ground return paths, and the analog-to-digital boundary need a deliberate layout review before routing begins. A continuous ground plane is preferred; “splitting ground” is not a substitute for controlling current-return paths.

4. MCU and codec implementation notes

4.1 STM32H573

The STM32H573 supplies the compute, real-time peripherals, USB interface, and development support. It is not being selected for its on-chip audio ADC/DAC; the external codec is the primary conversion path.

Required MCU support circuitry includes:

- all required VDD, VDDA, VDDUSB, ground, reset, and decoupling connections from the package-specific datasheet guidance;
- boot-configuration resistors and accessible reset;
- a dedicated SWD header with SWDIO, SWCLK, NRST, ground, and target-voltage reference;
- a clock plan that can generate the chosen audio clocks accurately;
- accessible test points for 3.3 V, reset, SWD, audio clocks, and key power rails.

The STM32H573 datasheet identifies separate VDD, VDDA, and VDDUSB domains. VDDUSB is specified as an independent 3.0 V to 3.6 V supply for the USB transceiver. These pins must be populated and decoupled exactly as required for the chosen package, even though audio conversion is external.

4.2 TLV320AIC3204

The TLV320AIC3204 is the board's audio interface. It combines the ADC/DAC functions with analog input and output routing and is controlled by I2C. The board will expose the following MCU-to-codec signals:

Signal group	Direction / purpose
MCLK, BCLK, WCLK	STM32-generated audio timing to codec
MCU TX → codec DIN	Processed samples sent to the codec DAC
codec DOUT → MCU RX	ADC samples sent to the STM32
I2C SCL, SDA	Register configuration and status control
CODEC RESET	Deterministic reset and bring-up

The codec supply implementation must follow the selected AIC3204 power mode, not a generic “3.3 V everywhere” assumption. Its data sheet defines separate IOVDD, LDOIN/headphone supply, AVDD, and internal digital-supply arrangements. The current plan is to use the codec's intended internal-LDO arrangement, with 3.3 V supplied to the appropriate input rails and local filtering/decoupling, while verifying the required AVDD and DVDD connections against the final power-mode diagram. I2C pull-ups are provisionally 4.7 kohm to the chosen logic rail; final value depends on bus capacitance and voltage level.

5. USB and debug

USB-C on Revision A has two jobs: a convenient physical data/programming connection and an optional low-power input source. It is not yet a commitment to USB Audio Class support.

The STM32 USB 2.0 full-speed device interface connects to the USB-C receptacle through:

- CC pull-down resistors that identify the board as a USB-C sink/device;
- low-capacitance USB ESD protection close to the connector;
- short, controlled USB D+ and D– routing;
- VBUS sensing and a power path that prevents unwanted back-feeding; and
- the MCU's correctly supplied VDDUSB domain.

SWD remains mandatory even when USB programming works. It gives a dependable recovery path for first bring-up, clock/configuration failures, boot configuration mistakes, and source-level debugging.

6. Power architecture

The board is intended to work from either a nominal 9 V pedal supply or USB-C 5 V without either source back-feeding the other. The current architecture uses a TPS2121 power mux to select between sources, then regulates to a stable internal system rail before generating logic and analog rails.

6.1 Source selection and front-end protection

The TPS2121 is a dual-input priority power mux with seamless source transition. Its operating range supports the intended 5 V USB and nominal 9 V pedal sources. Source priority, current-limit setting, UVLO behavior, connector polarity, and the exact treatment of USB VBUS must be shown explicitly on the schematic rather than inferred from net names.

The pedal-power input needs reverse-polarity protection and transient handling before it reaches the mux. The selected P-channel MOSFET approach is suitable for low-loss reverse-polarity protection, but its orientation, VGS limits, gate protection, inrush behavior, and thermal margin remain schematic-review items.

6.2 Regulation strategy

The TPS63070 accepts a 2.0 V to 16 V input and can buck or boost, making it appropriate for producing a stable intermediate rail when the input may be 5 V or 9 V. The present concept is a regulated 9V_SYS rail.

The TPS54202H is a 4.5 V to 28 V, 2 A synchronous buck regulator and is the candidate for the efficient downstream logic rail. The TLV755P is then reserved for local low-noise regulation where it is justified, rather than asking one regulator to serve every rail equally well.

This division is intentional: switching regulators handle the larger voltage conversion efficiently, while local filtering and a low-noise LDO support sensitive analog/codec rail requirements. Actual component values, compensation, inductor selection, power dissipation, startup sequencing, and worst-case current budget are not yet validated and must not be treated as final from this document.

6.3 Initial rail budget and validation

Before PCB release, create a rail-current estimate for the MCU at its intended clock rate, codec operating modes, op-amps, LEDs, USB load, and expansion headers. Then validate:

- no USB-to-pedal or pedal-to-USB back-feed under any plug/unplug order;
- stable source switchover without MCU brownout or codec reset anomalies;
- correct startup sequence and reset behavior;
- switching ripple and spurs at the codec/analog rails;
- regulator and MOSFET temperatures at maximum expected load; and
- adequate current and voltage margin for future peripherals.

7. Controls and expansion

Revision A includes four potentiometer inputs, at least two digital inputs, and at least three LEDs. These controls should be placed and labeled so a bench test can exercise them without an

enclosure. Potentiometers should be routed to MCU ADC-capable pins with reasonable source impedance and local filtering; switches should use defined pull-up/pull-down behavior and firmware debouncing.

Expansion headers should expose spare GPIO and practical I2C, SPI, and UART access with power and ground pins. The initial layout should not spend every convenient timer/DMA-capable pin before the audio, USB, SWD, and user-control functions are verified.

8. Bring-up plan

The board should be brought up in a sequence that limits the number of unknowns at each stage.

1. Inspect before power. Check assembly polarity, power-net resistance to ground, connector pinout, and rail isolation.
2. Power path only. Validate pedal input, USB input, mux selection, back-feed protection, and each regulator without the MCU running.
3. MCU core. Confirm reset, SWD connection, boot behavior, clock configuration, and a simple LED firmware image.
4. USB. Validate USB enumeration or a minimal device connection, VBUS sense, and recovery through SWD.
5. Codec control. Confirm I2C address response, reset sequence, supply voltages, and codec register read/write behavior.
6. Digital audio. Use a known clock and DMA configuration; first verify received ADC samples and transmitted DAC samples with a test tone or loopback.
7. Analog audio. Measure input gain, noise floor, clipping point, frequency response, output level, and audible pass-through quality.
8. DSP features. Add bypass, gain, clipping/overdrive, delay, filtering, or other effects one at a time with regression checks.

9. Known open decisions and risks

Topic	Current position	Closure criterion
Exact STM32 ordering code and pin map	STM32H573 family selected	Schematic pin assignment, memory needs, package availability, and PCB escape reviewed
Codec rail topology	Internal-LDO-oriented plan	Final AIC3204 power-mode schematic matches data-sheet requirements
Analog input buffer	Functional requirements established	Op-amp, bias, gain, protection capacitance, and headroom simulated/measured
Output format and level	Development output first	Measured level/noise and intended pedal integration defined
Audio clocks	STM32 master, codec slave	Stable 48 kHz I2S transfer with verified MCLK/BCLK/WCLK timing
USB role	Device/programming/communications first	USB descriptor and power-path behavior validated; USB audio deferred
9V_SYS and regulator values	Architecture selected	Current budget, compensation, ripple, thermal, and transient behavior validated
PCB layout	Four layers preferred	Placement and return-current review completed before routing sign-off

11. References

These are the design references used for the initial architecture. Store the licensed PDFs in references/datasheets/ if they are to be retained with the repository, or link to the manufacturers' current document pages instead.

- STMicroelectronics, STM32H573AI Data Sheet.
- STMicroelectronics, AN5711 Getting Started with STM32H5 MCU Hardware Development.
- STMicroelectronics, STM32H5 Reference Manual.
- Texas Instruments, TLV320AIC3204 Ultra Low Power Stereo Audio Codec Data Sheet.
- Texas Instruments, SLAA557 TLV320AIC3204 Application Reference Guide.
- Texas Instruments, TPS2121 Priority Power MUX Data Sheet.
- Texas Instruments, TPS63070 Buck Boost Converter Data Sheet.
- Texas Instruments, TPS54202H Synchronous Buck Converter Data Sheet.
- Texas Instruments, TLV755P Low Dropout Regulator Data Sheet.
- AOS, A03401A P Channel MOSFET Data Sheet.